Diseases and immunity

20 questions: 10 multiple choice, 10 structured.

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1 · Part 1 Core Diseases and immunity [1]

Which statement describes a transmissible disease?

A a disease caused by a lack of one nutrient in the diet

B a disease in which the pathogen can be passed from one host to another

C a disease that is passed from parents to their offspring in their genes

D a disease that develops because body cells stop working properly with age

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2 · Part 1 Core Diseases and immunity [1]

Which example shows a pathogen being transmitted indirectly?

A a nurse is infected when a patient's blood gets into a cut on her hand

B a child is infected after kissing a family member who has a cold sore

C a baby is infected by body fluids from its mother during birth

D an office worker touches a door handle used by an infected colleague, then touches his mouth

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3 · Part 1 Core Diseases and immunity [1]

Which row correctly describes how mucus and stomach acid defend the body against pathogens?

	mucus	stomach acid
A	traps pathogens	kills pathogens
B	kills pathogens	traps pathogens
C	traps pathogens	produces antibodies
D	produces antibodies	kills pathogens

4 · Part 1 Supplement Diseases and immunity [1]

Antibodies made against the measles virus do not bind to the chickenpox virus.

What is the reason for this?

A antibodies are only made against bacteria, not against viruses

B the chickenpox virus has no antigens on its surface

C the antibodies have a shape complementary to measles antigens but not to chickenpox antigens

D the antibodies are destroyed by phagocytes before they reach the chickenpox virus

5 · Part 1 Supplement Diseases and immunity [1]

Which statement describes antibodies?

A proteins that bind to antigens, leading to direct destruction of pathogens or marking of pathogens for destruction by phagocytes

B cells that engulf and digest pathogens

C carbohydrates that coat pathogens so that they cannot reproduce

D proteins that bind to any pathogen, whatever the shape of its antigens

6 · Part 1 Supplement Diseases and immunity [1]

What is put into the body during vaccination?

A antibodies made by another person

B memory cells

C white blood cells

D weakened pathogens or their antigens

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7 · Part 1 Supplement Diseases and immunity [1]

Which row is correct for passive immunity?

	antibodies made by the person's own lymphocytes	memory cells produced	length of protection
A	yes	yes	long-term
B	no	no	short-term
C	no	yes	long-term
D	yes	no	short-term

8 · Part 1 Supplement Diseases and immunity [1]

Which person has gained active immunity?

A a baby who receives antibodies in breast milk

B a fetus that receives antibodies across the placenta

C an adult who recovered from an infection and produced antibodies against the pathogen

D a patient who is injected with antibodies taken from the blood of another person

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9 · Part 1 Supplement Diseases and immunity [1]

The list shows some events that happen after a person drinks water containing the cholera bacterium.

1 water moves into the gut by osmosis

2 the bacterium produces a toxin

3 chloride ions are secreted into the small intestine

4 diarrhoea causes dehydration

What is the correct order of these events?

A 2 → 1 → 3 → 4

B 3 → 2 → 1 → 4

C 2 → 3 → 4 → 1

D 2 → 3 → 1 → 4

10 · Part 1 Supplement Diseases and immunity [1]

Which row describes cholera?

	type of pathogen	main way it is transmitted
A	bacterium	in contaminated water
B	virus	in contaminated water
C	bacterium	in droplets in the air
D	fungus	in contaminated water

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11 · Part 1 Core Diseases and immunity [6]

After a large flood, the water pipes and sewers in a town were damaged. Rubbish was not collected for several weeks. Many people then became ill with a disease that causes diarrhoea. The pathogen was found in the drinking water and in piles of uncollected rubbish.

(a) The disease is a transmissible disease. State what is meant by a transmissible disease. [1]

(b) Drinking contaminated water is one indirect way in which a pathogen can be transmitted. State two other indirect ways in which pathogens can be transmitted. [2]

(c) Explain why repairing the sewage treatment system and collecting the rubbish would help to control the spread of the disease. [3]

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12 · Part 1 Core Diseases and immunity [5]

The human body has several defences against pathogens.

(a) Describe how hairs in the nose and mucus help to stop pathogens in the air from entering the lungs. [2]

(b) Explain why a deep cut in the skin increases the risk of infection. [1]

(c) Some pathogens get past these defences and enter the blood. Describe how white blood cells defend the body against these pathogens. [2]

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13 · Part 1 Core Diseases and immunity [7]

The manager of a school canteen recorded the number of students with food poisoning in two terms.

At the start of term 2, two new rules were introduced for canteen staff:

- rule 1: wash hands with soap after using the toilet and before handling food
- rule 2: store and prepare raw meat separately from cooked food

The table shows the results.

term	**number of students eating in the canteen**	**number of students with food poisoning**
term 1 (before the rules)	1200	48
term 2 (after the rules)	1250	15

(a) Calculate the percentage of students eating in the canteen who had food poisoning in each term. [2]

(b) Explain why it is better to compare these percentages than to compare the numbers of students with food poisoning. [1]

(c) Explain how each of the two rules helps to reduce food poisoning. [4]

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14 · Part 1 Supplement Diseases and immunity [7]

A person was given a first dose of a vaccine on day 0 and a second dose of the same vaccine on day 60. The table shows the concentration of antibodies against the pathogen in the person's blood.

time after first dose / days	antibody concentration / arbitrary units
0	0
10	15
20	30
30	20
40	10
60	5
70	160
80	240
100	200

(a) State what is put into the body in a vaccine. [1]

(b) Outline how the first dose of the vaccine leads to an increase in antibody concentration between day 0 and day 20. [2]

(c) Use the data to compare the response to the second dose with the response to the first dose. [2]

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(d) Explain why the response to the second dose is different from the response to the first dose. [2]

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15 · Part 1 Supplement Diseases and immunity [6]

The table shows the percentage of children vaccinated against measles and the number of measles cases in one country over 15 years.

year	percentage of children vaccinated	number of measles cases
2000	45	24 000
2005	60	15 600
2010	80	4800
2015	95	360

(a) Describe the relationship between the percentage of children vaccinated and the number of measles cases. [1]

(b) Calculate the percentage decrease in the number of measles cases between 2000 and 2015. [2]

(c) Explain how vaccinating a large percentage of the population reduces the spread of measles, including to people who have not been vaccinated. [3]

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16 · Part 1 Supplement Diseases and immunity [6]

Newborn babies are protected against some pathogens by antibodies from their mother.

(a) State how antibodies pass from the mother to the fetus before birth. [1]

(b) Explain the importance of breast-feeding for the development of passive immunity in a young baby. [3]

(c) The protection given by passive immunity lasts only a short time. Explain why. [2]

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17 · Part 1 Supplement Diseases and immunity [7]

Cholera is a disease that causes severe diarrhoea.

(a) State the type of organism that causes cholera and how cholera is transmitted. [2]

(b) Explain how the cholera pathogen causes diarrhoea. [3]

(c) The cholera pathogen stays inside the gut. Explain why a person with cholera can still become dangerously ill. [2]

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18 · Part 1 Supplement Diseases and immunity [5]

Two strains of a bacterium, strain X and strain Y, cause the same type of illness. Strain X and strain Y have different antigens.

One woman recovered from an infection by strain X. One year later she was infected by strain Y and became ill again.

(a) State how the antigens of strain X differ from the antigens of strain Y. [1]

(b) Explain why the antibodies that the woman made against strain X did not protect her against strain Y. [2]

(c) Describe two ways in which antibodies that bind to antigens lead to the destruction of a pathogen. [2]

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19 · Part 1 Supplement Diseases and immunity [6]

Two scientists are going to work for one year in a region where a viral disease, disease P, is common.

- Scientist 1 is vaccinated against disease P eight weeks before travelling.
- Scientist 2 has no time to be vaccinated. The day before travelling, she is injected with antibodies against the disease P virus that were collected from the blood of another person.

(a) State the type of immunity each scientist has against disease P when they arrive. [2]

(b) Predict which scientist is more likely to still be protected against disease P at the end of the year. Explain your answer. [4]

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20 · Part 1 Supplement Diseases and immunity [7]

There was an outbreak of cholera in a city. The city has two districts:

- district K gets its drinking water from a treated, piped supply, and its sewage is treated
- district L gets its drinking water from a river, and untreated sewage flows into the river upstream of the district

The table shows the number of cholera cases in each district during the outbreak.

district	population	number of cholera cases
K	40 000	60
L	25 000	900

(a) Calculate the number of cholera cases per 1000 people in each district. [2]

(b) Use the information to explain the difference in the number of cases per 1000 people in the two districts. [3]

(c) Health officials plan to vaccinate people in district L against cholera. Suggest why they should also provide district L with a clean water supply. [2]

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